

DAILY DRILLING REPORT SET

SYN-Assam-Duliajan-003 (W003) — SYN-Assam-Duliajan

1. REPORT SET IDENTIFICATION

Well ID	W003
Well Name	SYN-Assam-Duliajan-003
Field / Cluster	SYN-Assam-Duliajan
Area	Synthetic Assam Field Cluster (Duliajan-type operating area)
Latitude / Longitude	27.329164 N / 95.330380 E
Rig	SYN-RIG-E2000-07
Spud Date	10-Dec-2018
Final Report Date	22-Dec-2018
Daily Reports in Set	13 (Day 01 to Day 13)
Total Depth Reached	3,850 m
Primary Formation	Sandstone
Events Reported in Set	8

2. DEPTH PROGRESSION SUMMARY

Day	Date	From (m)	To (m)	Drilled (m)	Formation	Events
01	10-Dec-2018	0	565	565	Claystone	—
02	11-Dec-2018	565	1,005	440	Claystone / Shale	Casing Issue, Casing Issue
03	12-Dec-2018	1,005	1,250	245	Shale	Casing Issue, Casing Issue
04	13-Dec-2018	1,250	1,560	310	Shale	Stuck Pipe
05	14-Dec-2018	1,560	1,925	365	Shale	—
06	15-Dec-2018	1,925	2,260	335	Shale / Sandstone	Mud Loss
07	16-Dec-2018	2,260	2,500	240	Sandstone	—
08	17-Dec-2018	2,500	2,785	285	Sandstone	—
09	18-Dec-2018	2,785	2,960	175	Sandstone	Mud Loss
10	19-Dec-2018	2,960	3,180	220	Sandstone	Cementing Issue
11	20-Dec-2018	3,180	3,415	235	Sandstone	—
12	21-Dec-2018	3,415	3,640	225	Sandstone / Limestone	—
13	22-Dec-2018	3,640	3,850	210	Limestone	—

DAILY DRILLING REPORT — DAY 01	W003 · 10-Dec-2018
Well ID: W003 Well Name: SYN-Assam-Duliajan-003 Report Date: 2018-12-10 Day Number: 1 Start Depth: 0 m End Depth: 565 m Drilled Interval: 565 m Formation: Claystone Mud Weight: 9.42 ppg ROP: 32.7 m/hr RPM: 140 WOB: 7.1 klbf Events: None NPT: 0.0 hours	
Operations performed. - Rig up completed and well spudded at 10-Dec-2018. Drilled 17-1/2" hole from surface to 565 m. - Slipped and cut drilling line; routine BOP function test carried out.	
Geological observations. Cuttings throughout the interval are claystone, consistent with the 0–867 m unit logged in the offset wells of this cluster.	
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 9.42 ppg, properties held within programme; no losses or gains recorded.	
Equipment / operational observations. Solids-control equipment cleaned; shaker screens changed. No downtime.	
Events reported this day. None. Operations proceeded within the planned parameter envelope.	
End-of-day status. Depth 565 m in Claystone. Drilling ahead. No non-productive time this tour.	

DAILY DRILLING REPORT — DAY 02

W003 · 11-Dec-2018

Well ID: W003

Well Name: SYN-Assam-Duliajan-003

Report Date: 2018-12-11

Day Number: 2

Start Depth: 565 m

End Depth: 1005 m

Drilled Interval: 440 m

Formation: Claystone / Shale

Mud Weight: 10.15 ppg

ROP: 25.7 m/hr

RPM: 136

WOB: 11.6 klbf

Events: Casing Issue at 612 m (Low); Casing Issue at 958 m (Low)

NPT: 12.2 hours

Operations performed.

• Drilled 440 m of section from 565 m to 1,005 m (4.3 hr on bottom).

• Circulated and conditioned mud; hi-vis sweep circulated bottoms-up.

• Pulled out of hole, ran 13-3/8" casing to 865 m and cemented as programmed (Lead 12.6 ppg / Tail 15.8 ppg, cement to surface). Waited on cement and pressure tested.

• Casing Issue handled at 612 m — 6.6 hr non-productive time. Casing run suspended, string pulled back and the hole conditioned/reamed over the restriction before the casing was rerun; pressure test repeated after landing.

• Casing Issue handled at 958 m — 5.6 hr non-productive time. Casing run suspended, string pulled back and the hole conditioned/reamed over the restriction before the casing was rerun; pressure test repeated after landing.

Geological observations.

Formation change crossed during the tour: Claystone to 867 m, Shale to 2,190 m. Cuttings confirm the tops used for correlation with adjacent wells in the cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 10.15 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Casing/cementing equipment inspected after the event; cement unit and pressure recorders checked and found serviceable.

Events reported this day.

Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact
E0030	612	Claystone	Casing	Low	6.6	Minor non-productive time; operations resumed after routine parameter adjustment.

Prototype Training Record · Synthetic demonstration data

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DAILY DRILLING REPORT — DAY 03						W003 · 12-Dec-2018
Well ID: W003 Well Name: SYN-Assam-Duliajan-003 Report Date: 2018-12-12 Day Number: 3 Start Depth: 1005 m End Depth: 1250 m Drilled Interval: 245 m Formation: Shale Mud Weight: 10.71 ppg ROP: 17.1 m/hr RPM: 128 WOB: 15.5 klbf Events: Casing Issue at 1049 m (Medium); Casing Issue at 1169 m (Critical) NPT: 59.8 hours						
Operations performed. - Drilled 245 m of Shale from 1,005 m to 1,250 m (2.0 hr on bottom). - Circulated and conditioned mud; hi-vis sweep circulated bottoms-up. - Casing Issue handled at 1,049 m — 14.2 hr non-productive time. Casing run suspended, string pulled back and the hole conditioned/reamed over the restriction before the casing was rerun; pressure test repeated after landing. - Casing Issue handled at 1,169 m — 45.6 hr non-productive time. Casing run suspended, string pulled back and the hole conditioned/reamed over the restriction before the casing was rerun; pressure test repeated after landing.						
Geological observations. Cuttings throughout the interval are shale, consistent with the 867–2,190 m unit logged in the offset wells of this cluster.						
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 10.71 ppg, properties held within programme; no losses or gains recorded.						
Equipment / operational observations. Casing/cementing equipment inspected after the event; cement unit and pressure recorders checked and found serviceable.						
Events reported this day.						
Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact
E0031	1,049	Shale	Casing Issue	Medium	14.2	Partial non-productive time; drilling suspended while the well was treated and conditioned.
E0033	1,169	Shale	Casing Issue	Critical	45.6	Major non-productive time; section objectives affected and specialist intervention required.
End-of-day status. Depth 1,250 m in Shale. Drilling ahead. Non-productive time this tour 59.8 hr.						

DAILY DRILLING REPORT — DAY 04						W003 · 13-Dec-2018
Well ID: W003 Well Name: SYN-Assam-Duliajan-003 Report Date: 2018-12-13 Day Number: 4 Start Depth: 1250 m End Depth: 1560 m Drilled Interval: 310 m Formation: Shale Mud Weight: 11.10 ppg ROP: 15.8 m/hr RPM: 125 WOB: 18.2 klbf Events: Stuck Pipe at 1447 m (Medium) NPT: 14.1 hours						
Operations performed. - Drilled 310 m of Shale from 1,250 m to 1,560 m (8.4 hr on bottom). - Circulated and conditioned mud; hi-vis sweep circulated bottoms-up. - Stuck Pipe handled at 1,447 m — 14.1 hr non-productive time. String worked and jarred within the allowable overpull; lubricant pill spotted and hole cleaning improved with high-viscosity sweeps and increased rotation before back-reaming.						
Geological observations. Cuttings throughout the interval are shale, consistent with the 867–2,190 m unit logged in the offset wells of this cluster.						
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 11.10 ppg, properties held within programme; no losses or gains recorded.						
Equipment / operational observations.						

DAILY DRILLING REPORT — DAY 05

W003 · 14-Dec-2018

Well ID: W003
Well Name: SYN-Assam-Duliajan-003
Report Date: 2018-12-14
Day Number: 5
Start Depth: 1560 m
End Depth: 1925 m
Drilled Interval: 365 m
Formation: Shale
Mud Weight: 11.58 ppg
ROP: 15.2 m/hr
RPM: 120
WOB: 20.2 klbf
Events: None
NPT: 0.0 hours

Operations performed.

• Drilled 365 m of Shale from 1,560 m to 1,925 m (22.5 hr on bottom).

• Deviation survey taken; hole angle within tolerance.

Geological observations.

Cuttings throughout the interval are shale, consistent with the 867–2,190 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 11.58 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

All rig equipment functioning normally; no downtime recorded.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 1,925 m in Shale. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 06

W003 · 15-Dec-2018

Well ID: W003
Well Name: SYN-Assam-Duliajan-003
Report Date: 2018-12-15
Day Number: 6
Start Depth: 1925 m
End Depth: 2260 m
Drilled Interval: 335 m
Formation: Shale / Sandstone
Mud Weight: 12.02 ppg
ROP: 15.9 m/hr
RPM: 117
WOB: 22.6 klbf
Events: Mud Loss at 2210 m (Medium)
NPT: 6.2 hours

Operations performed.

• Drilled 335 m of section from 1,925 m to 2,260 m (10.3 hr on bottom).

• Slipped and cut drilling line; routine BOP function test carried out.

• Pulled out of hole, ran 9-5/8" casing to 2,190 m and cemented as programmed (Lead 12.9 ppg / Tail 15.8 ppg, TOC 750 m). Waited on cement and pressure tested.

• Mud Loss handled at 2,210 m — 6.2 hr non-productive time. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.

Geological observations.

Formation change crossed during the tour: Shale to 2,190 m, Sandstone to 3,427 m. Cuttings confirm the tops used for correlation with adjacent wells in the cluster.

Mud / circulation observations.

Returns reduced during the tour; losses treated with LCM and mud weight trimmed. Loss volumes recorded on the daily loss sheet. Mud weight held at 12.02 ppg after treatment.

Equipment / operational observations.

Mud pump No. 2 packing changed during routine circulation; no lost time.

Events reported this day.

Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact
E0028	2,210	Sandstone	Mud Loss	Medium	6.2	Partial non-productive time; drilling suspended while the well was treated and conditioned.

Prototype Training Record · Synthetic demonstration data

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DAILY DRILLING REPORT — DAY 07

W003 · 16-Dec-2018

Well ID: W003
Well Name: SYN-Assam-Duliajan-003
Report Date: 2018-12-16
Day Number: 7
Start Depth: 2260 m
End Depth: 2500 m
Drilled Interval: 240 m
Formation: Sandstone
Mud Weight: 12.33 ppg
ROP: 16.3 m/hr
RPM: 114
WOB: 24.0 klb
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 240 m of Sandstone from 2,260 m to 2,500 m (22.5 hr on bottom).
- Made connections and circulated bottoms-up at intervals; shakers monitored.

Geological observations.

Cuttings throughout the interval are sandstone, consistent with the 2,190–3,427 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 12.33 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

All rig equipment functioning normally; no downtime recorded.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 2,500 m in Sandstone. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 08

W003 · 17-Dec-2018

Well ID: W003
Well Name: SYN-Assam-Duliajan-003
Report Date: 2018-12-17
Day Number: 8
Start Depth: 2500 m
End Depth: 2785 m
Drilled Interval: 285 m
Formation: Sandstone
Mud Weight: 12.62 ppg
ROP: 15.0 m/hr
RPM: 109
WOB: 25.4 klb
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 285 m of Sandstone from 2,500 m to 2,785 m (22.5 hr on bottom).
- Made connections and circulated bottoms-up at intervals; shakers monitored.

Geological observations.

Cuttings throughout the interval are sandstone, consistent with the 2,190–3,427 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 12.62 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Solids-control equipment cleaned; shaker screens changed. No downtime.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 2,785 m in Sandstone. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 09						W003 · 18-Dec-2018
Well ID: W003 Well Name: SYN-Assam-Duliajan-003 Report Date: 2018-12-18 Day Number: 9 Start Depth: 2785 m End Depth: 2960 m Drilled Interval: 175 m Formation: Sandstone Mud Weight: 12.96 ppg ROP: 14.1 m/hr RPM: 108 WOB: 27.0 klbf Events: Mud Loss at 2840 m (High) NPT: 12.3 hours						
Operations performed. - Drilled 175 m of Sandstone from 2,785 m to 2,960 m (10.2 hr on bottom). - Slipped and cut drilling line; routine BOP function test carried out. - Mud Loss handled at 2,840 m — 12.3 hr non-productive time. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.						
Geological observations. Cuttings throughout the interval are sandstone, consistent with the 2,190–3,427 m unit logged in the offset wells of this cluster.						
Mud / circulation observations. Returns reduced during the tour; losses treated with LCM and mud weight trimmed. Loss volumes recorded on the daily loss sheet. Mud weight held at 12.96 ppg after treatment.						
Equipment / operational observations. All rig equipment functioning normally; no downtime recorded.						
Events reported this day.						
Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact
E0003	2,840	Sandstone	Mud Loss	High	12.3	Significant non-productive time; remedial operations and additional materials required.
End-of-day status. Depth 2,960 m in Sandstone. Drilling ahead. Non-productive time this tour 12.3 hr.						

DAILY DRILLING REPORT — DAY 10						W003 · 19-Dec-2018
Well ID: W003 Well Name: SYN-Assam-Duliajan-003 Report Date: 2018-12-19 Day Number: 10 Start Depth: 2960 m End Depth: 3180 m Drilled Interval: 220 m Formation: Sandstone Mud Weight: 13.25 ppg ROP: 13.6 m/hr RPM: 104 WOB: 29.5 klbf Events: Cementing Issue at 2987 m (Low) NPT: 3.6 hours						
Operations performed. - Drilled 220 m of Sandstone from 2,960 m to 3,180 m (18.9 hr on bottom). - Deviation survey taken; hole angle within tolerance. - Cementing Issue handled at 2,987 m — 3.6 hr non-productive time. Job evaluated against the pump/returns record; remedial squeeze / top-up cementing carried out and a cement bond log (CBL/VDL) run to confirm coverage.						
Geological observations. Cuttings throughout the interval are sandstone, consistent with the 2,190–3,427 m unit logged in the offset wells of this cluster.						
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 13.25 ppg, properties held within programme; no losses or gains recorded.						
Equipment / operational observations. Casing/cementing equipment inspected after the event; cement unit and pressure recorders checked and found serviceable.						
Events reported this day.						
Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact
Proteome Training Record - Synthetic demonstration data						

DAILY DRILLING REPORT — DAY 11

W003 · 20-Dec-2018

Well ID: W003
Well Name: SYN-Assam-Duliajan-003
Report Date: 2018-12-20
Day Number: 11
Start Depth: 3180 m
End Depth: 3415 m
Drilled Interval: 235 m
Formation: Sandstone
Mud Weight: 13.58 ppg
ROP: 12.9 m/hr
RPM: 104
WOB: 30.7 klbf
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 235 m of Sandstone from 3,180 m to 3,415 m (22.5 hr on bottom).
- Slipped and cut drilling line; routine BOP function test carried out.

Geological observations.

Cuttings throughout the interval are sandstone, consistent with the 2,190–3,427 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 13.58 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

All rig equipment functioning normally; no downtime recorded.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 3,415 m in Sandstone. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 12

W003 · 21-Dec-2018

Well ID: W003
Well Name: SYN-Assam-Duliajan-003
Report Date: 2018-12-21
Day Number: 12
Start Depth: 3415 m
End Depth: 3640 m
Drilled Interval: 225 m
Formation: Sandstone / Limestone
Mud Weight: 14.19 ppg
ROP: 7.9 m/hr
RPM: 88
WOB: 36.5 klbf
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 225 m of section from 3,415 m to 3,640 m (16.5 hr on bottom).
- Circulated and conditioned mud; hi-vis sweep circulated bottoms-up.
- Pulled out of hole, ran 7" casing to 3,610 m and cemented as programmed (Tail 15.8 ppg, TOC 2,040 m). Waited on cement and pressure tested.

Geological observations.

Formation change crossed during the tour: Sandstone to 3,427 m, Limestone to 3,850 m. Cuttings confirm the tops used for correlation with adjacent wells in the cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 14.19 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Mud pump No. 2 packing changed during routine circulation; no lost time.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 3,640 m in Limestone. 7" casing cemented at 3,610 m; waiting on cement.

DAILY DRILLING REPORT — DAY 13	W003 · 22-Dec-2018
Well ID: W003 Well Name: SYN-Assam-Duliajan-003 Report Date: 2018-12-22 Day Number: 13 Start Depth: 3640 m End Depth: 3850 m Drilled Interval: 210 m Formation: Limestone Mud Weight: 14.59 ppg ROP: 6.8 m/hr RPM: 85 WOB: 38.6 klbf Events: None NPT: 0.0 hours	
Operations performed. • Drilled 210 m of Limestone from 3,640 m to 3,850 m (16.5 hr on bottom). • Slipped and cut drilling line; routine BOP function test carried out. • Pulled out of hole, ran 5" liner casing to 3,850 m and cemented as programmed (Tail 15.8 ppg, liner lap tested). Waited on cement and pressure tested. • Reached driller's total depth of 3,850 m. Circulated hole clean, logged and secured the well. Geological observations. Cuttings throughout the interval are limestone, consistent with the 3,427–3,850 m unit logged in the offset wells of this cluster. Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 14.59 ppg, properties held within programme; no losses or gains recorded. Equipment / operational observations. Solids-control equipment cleaned; shaker screens changed. No downtime. Events reported this day. None. Operations proceeded within the planned parameter envelope. End-of-day status. Depth 3,850 m in Limestone. 5" liner casing cemented at 3,850 m; waiting on cement.	